UNDERWATER COMMUNICATION SYSTEM FOR SCUBA DIVERS

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1. Abstract

Underwater wireless communication (UWC) plays a critical role in applications such as tactical surveillance, marine biology monitoring, pollution detection, and underwater infrastructure maintenance. Traditional radio frequency (RF) communication methods, including Wi-Fi, are ineffective underwater due to rapid absorption of radio waves. In contrast, optical wireless communication (OWC) using light, particularly Li-Fi offers a promising alternative, as light can penetrate water more effectively. This paper presents a laser-based underwater communication system designed for real-time audio transmission. Pulse Amplitude Modulation (PAM) is employed to encode audio signals into laser pulses, which are transmitted through water and received by a solar panel functioning as a photodetector. A comprehensive literature review highlights the use of different laser wavelengths to optimize transmission efficiency, range, and error rates, and discusses acknowledgment mechanisms to detect signal loss. Experimental results demonstrate successful audio transmission in water over a 1-meter distance and text transmission in air over 5 meters using a line-of-sight setup, with minimal latency. Key challenges encountered include beam divergence, water turbidity, and alignment sensitivity. Despite range limitations typically reported in literature (7–10 meters underwater), our prototype illustrates the potential for low-cost, energy-efficient laser-based systems in applications such as diver communication, marine research, and underwater sensor networks.

2. Introductionn

Light Fidelity (Li-Fi) is a form of Visible Light Communication (VLC) technology developed by researchers at the University of Edinburgh. It is an emerging wireless communication method that enables high-speed data transmission using light-emitting diodes

(LEDs). Li-Fi operates by modulating the intensity of LED light at extremely high speeds—imperceptible to the human eye—to encode binary data (1s and 0s). Devices equipped with photodiodes can receive this modulated light and decode the transmitted information within the illumination area. Li-Fi offers several advantages over traditional Wi-Fi, including significantly lower latency, enhanced data throughput, improved security, and better accessibility in environments where radio frequency (RF) communication is limited. Under optimal conditions, Li-Fi has demonstrated data transmission speeds exceeding 1 Gbps. Additionally, since visible light cannot penetrate solid objects such as walls, Li-Fi naturally confines data transmission within a physical space, thereby offering enhanced data security. Furthermore, the data density of Li-Fi is up to 1000 times greater than that of Wi-Fi, making it a compelling solution for high-capacity wireless communication in densely populated or interference-prone environments.

Effective underwater communication is crucial for various applications, including marine research, search and rescue operations, environmental monitoring, and defence. Traditional communication methods such as acoustic and radio frequency (RF) transmission have limitations in underwater environments. Acoustic communication, while widely used, suffers from low data rates, latency, and environmental interference, whereas RF signals face high attenuation in water, restricting their range and efficiency. Wired communication, though

reliable, limits mobility and flexibility. To overcome these challenges, optical communication using lasers has emerged as a promising alternative due to its ability to achieve high-speed data transfer, minimal latency, and efficient energy consumption.

This project presents a laser-based underwater communication system designed to transmit real-time audio signals over short to medium distances. The system employs a laser transmitter to modulate audio signals using Pulse Amplitude Modulation (PAM). These modulated signals are transmitted through water and received by a solar panel, which acts as a photo-detector, converting the light pulses back into electrical signals. The received signals are then processed and amplified to produce audio output. Compared to other optical receivers, the use of a solar panel offers a cost-effective and power-efficient solution for underwater signal detection.

The effectiveness of laser-based communication depends on water clarity, as suspended particles and turbidity can cause scattering and absorption, affecting signal quality. Despite these challenges, laser communication offers several advantages over traditional methods, including high data transmission rates, narrow beam focus, and reduced interference, making it particularly suitable for applications in clear water environments.

This project aims to develop a reliable and efficient optical communication system tailored for scuba divers. Traditional diver communication methods, such as hand signals, are limited by visibility, while acoustic systems are often expensive and complex. By leveraging laser technology and optical sensors, this system enables light-based communication, enhancing underwater interactions and broadening the scope of underwater data transmission. The proposed system has significant potential applications, including diver-to-diver communication, underwater sensor networks, marine exploration, and defence operations, offering a practical and innovative solution for real-time underwater connectivity.

3. Literature Review

Various studies have explored underwater communication using optical and Li-Fi technologies. A paper presented a Li-Fi-based underwater communication system that faced challenges with signal attenuation and limited range, suggesting improvements in transmitter design,[1]. As demonstrated a prototype using a red laser and Arduino UNO for text transmission over 2 meters, achieving higher data rates than acoustic methods but suffering from line-of-sight limitations and noise, [9]. A visible light communication system for real-time audio transmission with a short range of 15 cm, suggesting the use of additional LED colors like blue or green for extended distance, [4]. A paper even explored a 5 Mbps optical wireless communication system with error correction coding, proposing a synchronization algorithm to ensure reliable data transmission,[3]. An underwater communication system for deep-sea divers using visible light, highlighting challenges with beam spread and recommending optimized transmitter designs. These studies collectively underscore the potential of optical communication for underwater applications, while identifying key areas for enhancement to achieve higher data rates, reliability, and extended communication range, [5]. The short transmission range and signal degradation in deep water, recommending advancements to enhance signal strength and mitigate environmental interference,[2].

4. Block Diagram

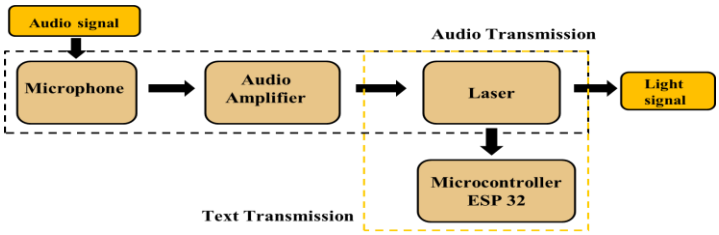


Fig. 1: Transmitter Block Diagram

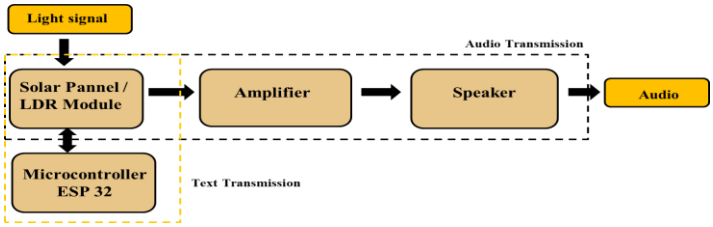


Fig. 2: Receiver Block Diagram

5. Methodology

The proposed underwater communication system for scuba divers follows a structured methodology encompassing system design, signal transmission and reception, and real-time implementation. This approach is developed to ensure efficient and reliable audio communication using laser-based technology in underwater environments.

The system consists of two main components: a transmitter and a receiver, both controlled by ESP32 microcontrollers for signal processing and overall coordination. On the transmitter side, audio signals are captured using an MAX4466 electret condenser microphone. These analog audio signals are then pre-processed and amplified using a PAM8403 module, which also facilitates pulse amplitude modulation (PAM). A red laser diode, is used to transmit the modulated audio signals through water using optical pulses.

For reception, the system departs from traditional methods by employing a solar panel instead of a Light Dependent Resistor (LDR) for detecting the laser light. The solar panel offers improved sensitivity and faster response times under varying underwater lighting conditions. When the modulated laser light reaches the receiver, the solar panel converts it directly into an electrical signal. This signal is then processed by the ESP32 microcontroller on the receiver side, which performs demodulation and converts the electrical signals back into an audio format. A PAM audio amplifier is subsequently used to enhance the demodulated audio signal, which is then output through a speaker for clear sound reproduction.

To ensure robust communication, an acknowledgment mechanism is implemented in the system. This mechanism allows the receiver to confirm successful reception of the signal. In case of a signal drop or error, a feedback signal is sent to the transmitter, prompting it to resend the lost data, thereby enhancing communication reliability.

The entire system is implemented and tested using the Arduino IDE, which allows real-time simulation and debugging. The accuracy and fidelity of the received audio signals are evaluated by comparing them with the original input. This comparison validates the effectiveness of the proposed underwater communication system using laser-based transmission and solar panel-based detection, highlighting its potential for practical deployment in underwater scenarios such as scuba diving communication.

Hardware Setup

The hardware implementation consists of cost-effective components optimized for real-time underwater communication:

1. **ESP32-WROOM 32** – Serves as the primary processing unit for both the transmitter and receiver, handling audio signal processing, modulation, and demodulation.
2. **MAX4466 Electret Condenser Microphone** – Captures audio input for transmission through the laser communication system.
3. **PAM8403 Module** – Amplifies and modulates the audio signal using pulse amplitude modulation (PAM) for efficient laser transmission.
4. **Laser Diode** – Emits a red laser beam carrying the modulated audio data, enabling light-based communication through water.
5. **Solar Panel** – Acts as the receiver sensor, detecting the transmitted laser signals and converting them into electrical signals with higher sensitivity and faster response than traditional LDRs.
6. **Speaker** – Outputs the reconstructed audio, enabling real-time communication for scuba divers.
7. **Power Supply and Supporting Modules** – Provide stable and reliable power to all components, ensuring continuous operation.

Software Setup

The system is designed to run on the Arduino IDE for programming and uploading code to the ESP32 modules. The software environment ensures efficient real-time audio transmission and reception using light-based communication.

Programming Environment

The Arduino IDE, along with the ESP32 board manager and required libraries, is used for coding and debugging. The implementation is carried out using C/C++ for low-level hardware interaction and real-time processing. Additional libraries such as SD.h for SD card access, SPI.h for communication, and driver/gpio.h for laser control are utilized.

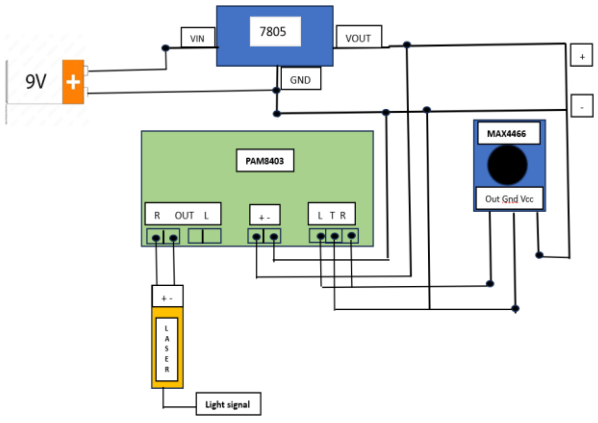


Fig. 3: Transmitter Circuit

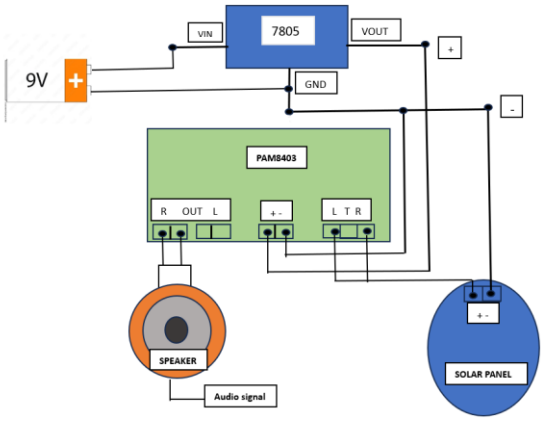


Fig. 4: Receiver Circuit

Technical Details

We used a red laser diode (650nm) with a 6mm size and 5V operating voltage, achieving a range of over 7-10 meters in the air. While blue (450nm) and green (520nm) lasers offer greater range in both air and water, they are costly and hazardous to humans. For demonstration purposes, the red laser was preferred, operating within 3V-5V. The ESP32-WROOM module, with an operating voltage of 2-3.6V, was selected for its versatile GPIO pins supporting functions like digital I/O, PWM, SPI, I²C, UART, and interrupts, along with capacitive touch sensing. It also features two 12-bit SAR ADCs with 18 channels and two 8-bit DACs for analog signal generation.

An Solar Pannel (3.3V-5V) was used for signal detection, with its sensitivity adjusted based on environmental conditions. The demonstration was conducted in a dark room to minimize interference, and a 1-meter-long tube was used as a controlled environment filled with either air or water. Arduino IDE was used to program the ESP32, implementing a basic algorithm to convert text into bits. The red laser transmits these bits via light signals, which are then detected and decoded using another ESP32 module paired with the LDR sensor.

The audio transmission system was implemented using a MAX4466 electret microphone module to capture analog audio signals. These signals were then amplified and modulated using a PAM8403 audio amplifier module. A 7805 voltage regulator was incorporated into the circuit to provide a stable 5V power supply to the components. The modulated audio signal was used to drive a laser diode, which served as the optical transmitter. The laser output was connected to the output terminal of the PAM8403 module, enabling real-time audio transmission via visible light.

For implementing acknowledgment functionality, both the laser transmitter and the microphone were interfaced with an ESP32 microcontroller. The ESP32 was programmed to detect the end of the audio signal and subsequently transmit a predefined binary sequence (e.g., 1010) via the laser to signal the end of transmission. On the receiver side, if the acknowledgment sequence is not received correctly or within the expected timeframe, a response is generated to indicate transmission failure or signal loss. This feedback mechanism enhances the reliability and integrity of the communication system by ensuring proper end-to-end data delivery.

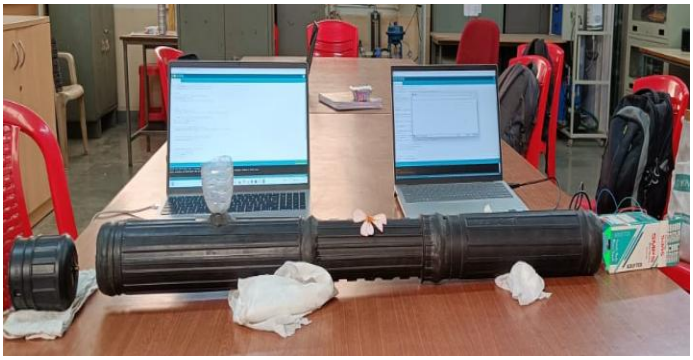


Fig. 5: Experimental Setup

6. Results

The results demonstrate that text can be successfully transmitted using light signals via a laser module, with no observed error rate. The experimental setup consisted of two ESP32 modules acting as a transmitter and receiver, a laser diode for signal transmission, and Solar panel for detection. In the air medium, the text was transmitted over a range of 5 meters, while in water, the range reached over 1 meter.

The experimental results highlight the importance of using filters and amplification circuits in laser-based audio transmission systems, as distortions and background noise were observed in the received audio output. The transmission was tested using a red laser over a distance of 1 meter in a controlled environment. Under optimal conditions, no significant loss of the audio signal was detected; however, a noticeable level of noise persisted in the output. It was observed that the signal quality deteriorated with an increase in transmission distance, with higher noise levels and reduced clarity. These findings emphasize the need for effective filtering and noise reduction techniques to enhance the quality and range of underwater or free-space optical audio communication.

Accurate text and audio reception was achieved without errors in the absence of obstacles, provided the laser and Solar panel were properly aligned. However, limitations

were observed in cases of misalignment or obstacles, leading to signal loss. These results validate the feasibility of using light signals for establishing a reliable communication system over significant distances in controlled environments.

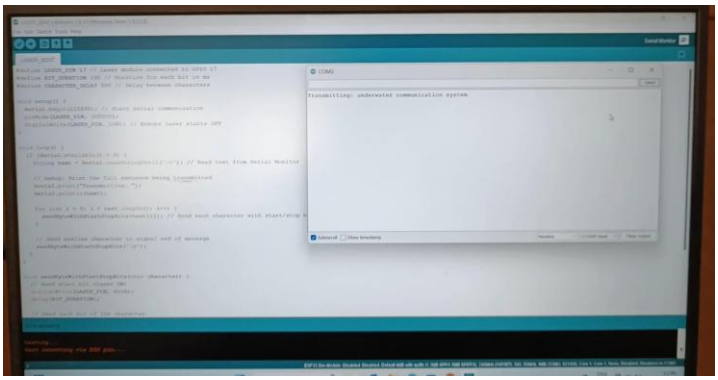


Fig. 6: Transmitting Text Signal

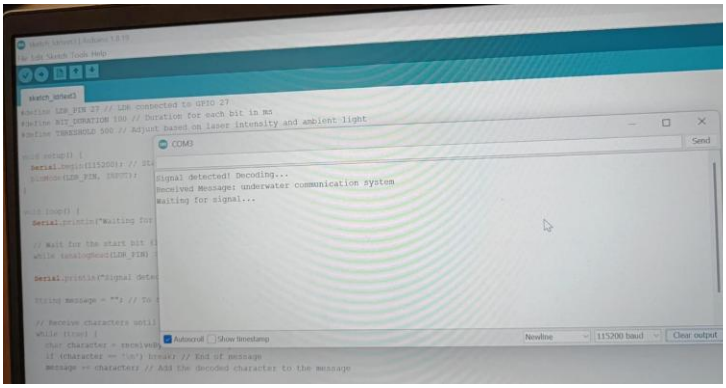


Fig. 7: Receiving Text Signal

7. Conclusion

The results of the underwater communication system demonstrate successful real-time audio transmission using laser light. The transmitter efficiently modulates audio signals using pulse amplitude modulation (PAM), and the laser diode emits the signals through the water. The solar panel at the receiver side accurately detects the laser signals, which are then demodulated and amplified to produce clear audio output through the speaker. The system maintained communication over short distances in controlled water environments, with minimal signal degradation. However, challenges such as beam divergence, water turbidity, and alignment issues were observed, which could affect long-range communication.

The study demonstrates that Li-Fi-based audio transmission is feasible but currently challenged by significant signal distortions and noise, necessitating the use of high-performance filtering and amplification techniques to ensure clear audio reception. Experimental observations suggest that employing higher-wavelength lasers, such as green or blue, can potentially extend the transmission range due to their better propagation characteristics in water. Accurate alignment between the laser source and the solar panel photodetector is also critical for minimizing data loss and maintaining reliable communication.
Furthermore, this research highlights the limited availability of comprehensive studies on underwater Li-Fi communication, indicating a need for further exploration and development in this field. With continued innovation and optimization, Li-Fi has the potential to evolve into a robust and efficient communication system suitable for various underwater and line-of-sight applications.

In conclusion, the proposed laser-based underwater communication system provides a cost-effective and reliable solution for short-distance audio transmission, making it suitable for applications like diver communication and underwater research, with potential for further optimization to enhance range and signal stability.

8. Future Scope

The future scope of the underwater communication system includes several potential enhancements to improve its performance and applicability. Advanced laser technologies, such as green or blue lasers, can be implemented to minimize signal attenuation and extend the communication range in murky water. Incorporating adaptive beam alignment systems using servo motors or sensors can ensure consistent laser targeting, reducing signal loss due to misalignment. Additionally, integrating error correction algorithms and signal processing techniques can enhance data accuracy and resilience against underwater disturbances. Further developments can involve the use of machine learning models for noise reduction and real-time signal analysis.

To enhance the functionality and reliability of the proposed Li-Fi communication system, future work can include the integration of an SD card module on the receiver side. This module would enable the storage of transmitted audio data, creating a permanent record of the communication. In the event of signal loss or transmission failure, the recorded data can be retransmitted automatically, eliminating the need for the user to repeat the message. Additionally, the incorporation of local storage facilitates the transfer of longer audio messages, extending the system's applicability to scenarios requiring delayed playback, archival, or batch communication. This enhancement

would significantly improve the system's usability and robustness in real-world applications, particularly in environments where continuous or real-time communication may be disrupted.

The system can also be expanded for multi-node communication, supporting larger networks for underwater exploration and surveillance. Moreover, combining this laser-based system with acoustic communication can provide a hybrid solution for long-range underwater data transmission. These advancements will significantly enhance the system's reliability, making it suitable for applications in marine research, underwater robotics, and military operations.

9. References

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